

SEMINAR TITLE

INTELLIGENT AI BASED CCTV SURVILANCE SYSTEM

A SYNOPSIS

Submitted for partial fulfillment for the Degree

of

Bachelor of Engineering

in

Computer Engineering

By

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1. Seminar Topic

Intelligent AI-Based CCTV Surveillance System for Real-Time Human Activity Detection, Tracking, and Safety Monitoring

2. Introduction / Background

Conventional CCTV surveillance systems primarily record video and require a human operator to continuously monitor multiple camera feeds. This approach becomes difficult to scale in crowded environments such as colleges, hostels, offices, public places, shopping areas, transportation facilities, and other sensitive locations. Continuous manual monitoring can lead to delayed detection of abnormal activities, operator fatigue, and missed events.

An Intelligent AI-Based CCTV Surveillance System uses computer vision and artificial intelligence to analyze live or recorded video automatically. The proposed system focuses on detecting people in video frames, assigning identities through multi-object tracking, extracting human pose/keypoints where applicable, and using activity-analysis techniques to identify potentially abnormal or unsafe behaviour such as fighting, physical aggression, crowding, or other predefined suspicious activities. The system can generate alerts and provide useful event information to a monitoring interface.

The topic combines concepts from Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Object Detection, Human Pose Estimation, Multi-Object Tracking, and real-time video processing. It also provides a practical foundation for extending the work into a final-year project and an industry-oriented surveillance prototype.

3. Motivation

The major motivation is to reduce dependence on continuous manual CCTV monitoring and improve the speed at which potentially dangerous events are identified. In real-world surveillance, the number of cameras can be large while the number of human operators is limited. An automated system can continuously inspect video streams and bring important events to the operator's attention.

4. Problem Statement

To design and develop an AI-based CCTV surveillance framework that processes live or recorded video, detects and tracks people, analyzes their movements and activities, identifies predefined abnormal or unsafe behaviours, and provides timely event information or alerts to assist security personnel in real-time monitoring.

5. Objectives

- Develop a system for detecting people in CCTV/video frames using modern computer-vision models.
- Track multiple detected persons across consecutive frames and maintain unique identities.
- Use human pose/keypoint information and activity-analysis techniques to understand human movement and predefined behaviours.
- Identify potentially abnormal activities such as fighting, physical aggression, crowding, or other selected unsafe events.
- Generate an understandable surveillance output/interface containing detection, tracking, confidence, and event information.

6. Scope of the Seminar

The seminar will study the architecture and implementation of an intelligent CCTV surveillance pipeline covering video acquisition, frame processing, person detection, multi-object tracking, human pose estimation/keypoint extraction, activity recognition, event classification, and alert generation.

The proposed scope includes live CCTV streams as well as recorded video, with a focus on people-centric surveillance. Suitable computer-vision datasets and benchmark video datasets may be considered for model evaluation. The system is intended as a prototype/research-oriented surveillance assistant rather than a replacement for human security personnel.

The initial scope does not claim universal recognition of every possible human activity, perfect identification in all environmental conditions, or autonomous law-enforcement decision making. Performance can be affected by camera angle, lighting, occlusion, video quality, crowd density, and dataset limitations.

7. Literature Survey

The literature survey should focus on recent and authoritative research in real-time object detection, multi-object tracking, human pose estimation, action/activity recognition, and abnormal-event detection in surveillance video. The following comparison identifies the major technical directions relevant to the proposed system. Specific paper details should be finalized from the papers actually selected and reviewed by the student.

Sr.	Author(s) & Year	Methodology	Dataset / Platform	Key Findings	Limitations	Future Scope
1	Wan et al. (2021)	Motion detection, superframe segmentation, feature fusion	SumMe, Hollywood2	Reduced redundant frames and improved long-video event retrieval.	Complex outdoor scenes; not focused on real-time person activity.	Real-time event detection and better activity understanding.
2	Watanabe et al. (2022)	Lightweight self-attention-based anomaly detection	UCF-Crime, ShanghaiTech, XD-Violence	Achieved efficient video anomaly detection with fewer parameters.	Limited representation of highly complex events.	Edge-based and real-time surveillance.
3	Zaheer et al. (2024)	Weakly supervised learning, clustering and normalcy suppression	UCF-Crime, ShanghaiTech, UCSD Ped2	Reduced dependency on frame-level anomaly labels.	Sensitive to noisy video-level labels.	Better anomaly localization and generalization.
4	Veesam & Satish (2024)	Enhanced person detection + Dynamic Mask Graph Network	Pedestrian Action Dataset	Improved person detection and continuous action prediction in CCTV scenes.	Computational complexity and occlusion issues.	Lightweight real-time multi-person activity recognition.
5	ICICV Authors (2025)	YOLOv8 + FaceNet/SVM + CNN-LSTM	Real-time surveillance videos	Combined human detection, tracking, identification and activity recognition.	Performance affected by crowded/complex scenes.	More robust real-time intelligent CCTV systems.

Recommended literature survey coverage: at least 5–8 recent peer-reviewed papers from IEEE Xplore, ACM Digital Library, Springer, ScienceDirect, or Google Scholar, with emphasis on comparative analysis rather than paper-by-paper description.

8. Research Gap / Problem Identification

Existing surveillance research provides strong individual solutions for object detection, tracking, pose estimation, and action recognition; however, a practical surveillance system must integrate these components while maintaining real-time performance. Important gaps include robust operation under occlusion and crowded scenes, reduction of false alarms, computational efficiency, reliable temporal activity recognition, and deployment on affordable hardware.

The proposed seminar therefore identifies the integration of person detection, identity tracking, pose/keypoint information, and behaviour analysis as a practical research and engineering challenge. The system can be further improved through better temporal models, domain-specific datasets, confidence calibration, edge deployment, and privacy-aware processing.

9. Technologies / Tools / Platforms

- Programming Language(s): Python
- Framework / Library: OpenCV, Ultralytics YOLO, NumPy, PyTorch, and other suitable computer-vision/deep-learning libraries
- Dataset(s): CCTV/surveillance video datasets and selected human-action/anomaly datasets appropriate to the activities being studied
- Hardware / Platform: CCTV/IP camera or recorded video source; GPU-enabled system preferred for real-time deep-learning inference
- Software / Tools: VS Code/PyCharm, Python environment, Git and GitHub, terminal, and optional web/dashboard framework
- Research / Reference Tools: IEEE Xplore

10. Expected Outcomes

- A working prototype capable of reading live or recorded CCTV/video input.
- Automatic detection of people with confidence information.
- Multi-person tracking with persistent IDs across video frames.
- Human pose/keypoint extraction for activity analysis where supported by the selected model.
- Identification of selected abnormal/unsafe activities based on the implemented activity-recognition approach.
- A monitoring output/dashboard that presents detections, tracking information and event alerts.
- Performance evaluation using suitable metrics such as precision, recall, F1-score, detection/tracking accuracy and inference speed, depending on the final implementation.

11. Relevance to Final Year Project / Future Scope

This seminar directly supports the development of the user's final-year CCTV Surveillance System by providing the theoretical and technical foundation for an end-to-end intelligent monitoring solution. The seminar concepts can be extended from a basic person-detection and tracking prototype to a complete real-time behavioural surveillance platform.

Future extensions may include real-time alerts through a web/mobile dashboard, multiple-camera support, crowd-density estimation, fall detection, intrusion detection, abandoned-object detection, improved fight/aggression recognition, audio-event analysis, edge-device deployment, cloud-based event storage, role-based access control, privacy-preserving processing, and model optimization for low-latency inference.

12. References

- [1] NVIDIA, OpenCV and PyTorch official documentation, as applicable to the final implementation.
- [2] Relevant recent IEEE Xplore Research papers
- [3] Relevant benchmark dataset papers and official dataset documentation used for evaluation.
- [4] Additional technical papers actually reviewed and cited during the seminar literature survey.

13. Approval

Student Signature: _____

Faculty Guide Signature: _____

Seminar Coordinator (As per division): _____

Date: _____